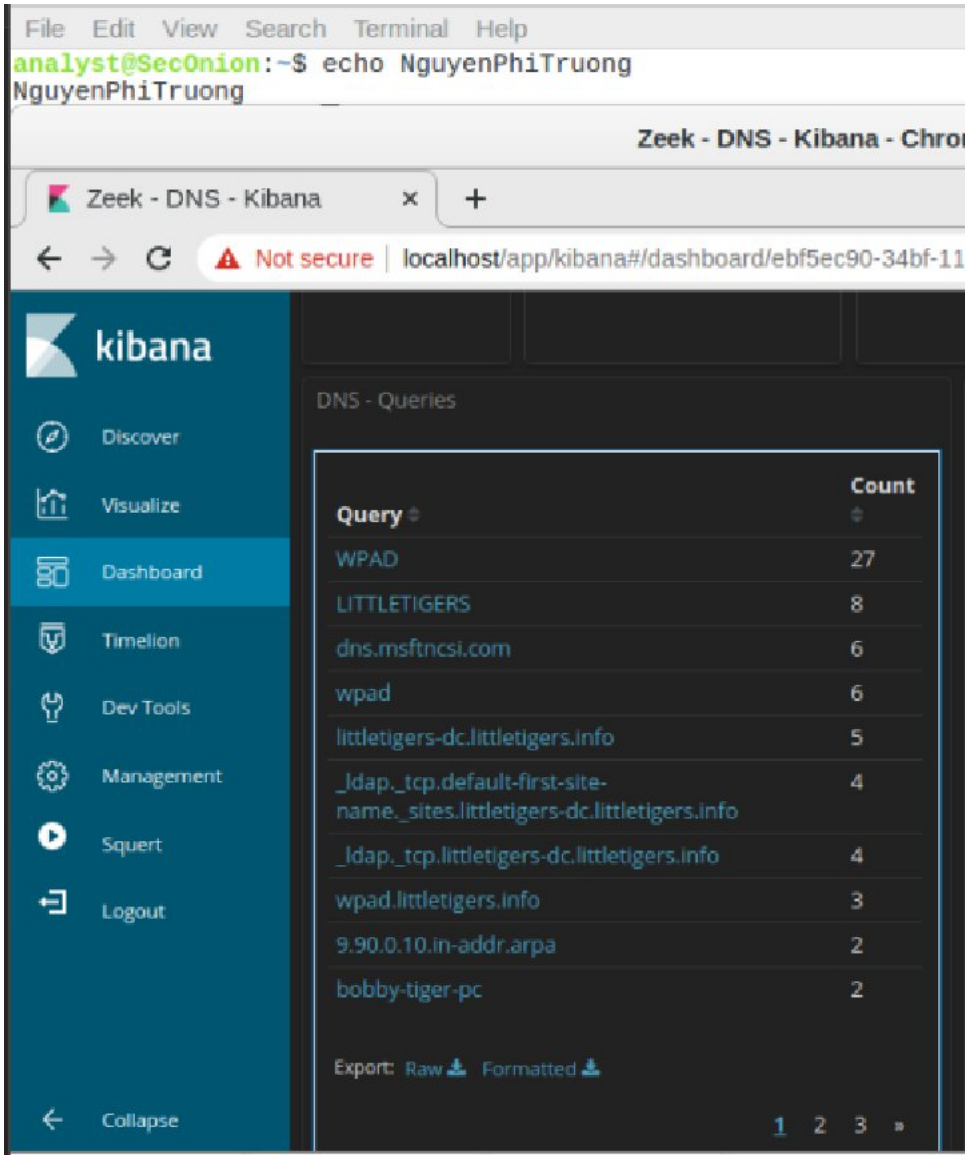
```
analyst@SecOnion:~$ echo NguyenPhiTruong
NguyenPhiTruong
analyst@SecOnion:~$ sha256sum test1.exe
2a9b0ed40f1f0bc0c13ff35d304689e9cadd633781cbcad1c2d2b92ced3f1c85  test1.exe
```

Calculates the SHA-256 cryptographic hash of the file test1.exe. This creates a unique string used to identify the file and check its reputation on malware analysis websites.



The background shows the "DNS - Queries" visualization in Kibana, displaying a list of domain names and the count of queries made by the infected Windows host (e.g., WPAD, LITTLETIGERS, www.msftncsi.com)

What safe procedures can be used to extract or generate a hash from a suspected malware file without risking infection?

To safely extract or generate a hash from a suspected malware file without risking infection, the following procedures should be used:

1. Use an Isolated Environment: Perform all extractions and analysis within an isolated Virtual Machine (like the Security Onion VM used in this lab) that is securely separated from your host computer and production network.
2. Do Not Execute the File: Never run, open, or double-click the suspected file.
3. Passive Extraction: Extract the file passively from network traffic captures (PCAP files) using tools like Wireshark (via the Export Objects feature) rather than downloading it directly to a normal operating system.
4. Use Command-Line Tools: Generate the hash using command-line utilities (such as sha256sum in a Linux terminal). This avoids graphical interface interactions that might accidentally trigger the execution of the malware.